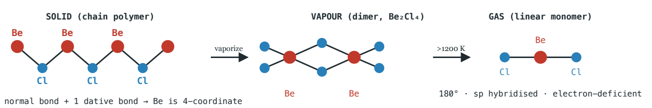
Alkaline Earth Metals — Group 2

NCERT + ALLEN + J.D. Lee · Structures & Reactions Companion

PART A — STRUCTURAL DIAGRAMS (VISUAL REFERENCE)

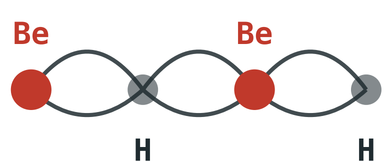
Key molecular and lattice structures referenced throughout the notes below.

4 · STRUCTURE — $BeCl_2$: THREE FACES OF ONE MOLECULE



Same molecule, three coordination states: 4-fold chain (solid) → 4-fold dimer (vapour) → 2-fold linear (gas, >1200 K)

5 · STRUCTURE — BeH_2 & SP-HYBRIDISATION



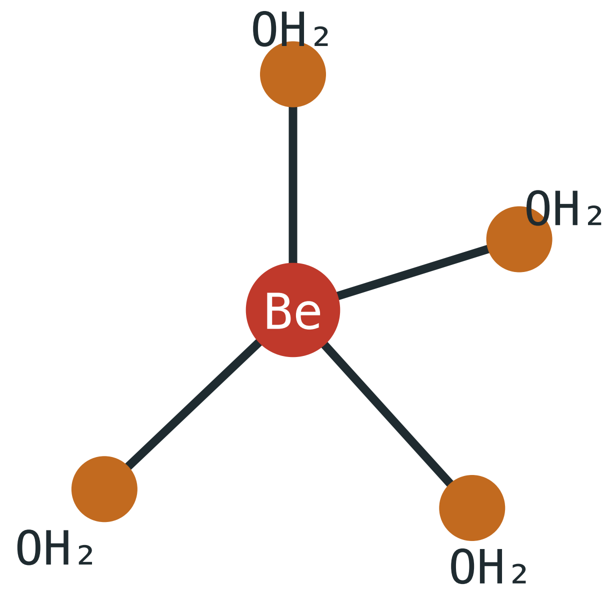
"banana" bonds — each H bridges 2 Be

Electron-deficient polymer: 2 valence e^- per Be spread over 4 bonds

MECHANISM Ground Be ($2s^2$) has 0 unpaired e^-. Promote 2s→2p: $2s^1 2p^1$, 2 unpaired e^- → **sp hybrid** → linear, 2-coordinate BeX_2 (gas only).

SYNTHESIS $2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$ — H^- (from $LiAlH_4$) attacks electrophilic Be, displacing Cl^-: an **SN2-like** substitution at Be. Purer route: $BeCl_2 + 2Li[BH_4] \rightarrow BeB_2H_8 + 2LiCl$, then $+2PPh_3 (\text{sealed}, \Delta) \rightarrow BeH_2 + 2Ph_3PBH_3$.

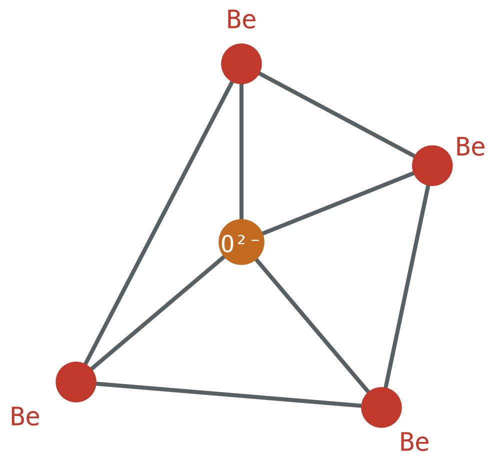
6 · STRUCTURE — Be^{2+} COMPLEXES (CN = 4 ALWAYS)



$[Be(H_2O)_4]^{2+}$ — sp^3, tetrahedral

- Be has only 4 valence orbitals (2s + 3×2p) — **never** 6-coordinate, unlike Mg–Ba (use d-orbitals).
- Strong Be–O bond weakens O–H → acidic hydrolysis: $[Be(H_2O)_4]^{2+} \rightleftharpoons [Be(H_2O)_3(OH)]^+ + H_3O^+$
- Alkali gives beryllate: $[Be(OH)_4]^{2-}$ · F^- gives $[BeF_4]^{2-}$

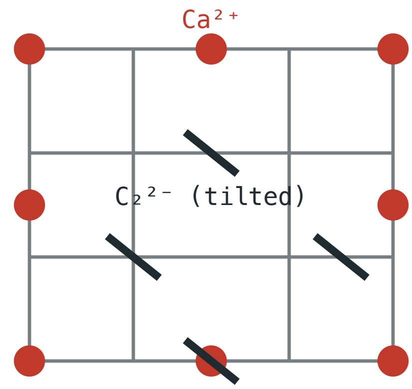
7 · STRUCTURE — BASIC BERYLLIUM ACETATE [$Be_4O(CH_3COO)_6$]



4 Be at tetrahedron corners; central O^{2-}; 6 acetates bridge the 6 edges

- Covalent, low-melting (285 °C), soluble in organic solvents — can be **distilled**, used to purify beryllium.
- Prepared by evaporating $Be(OH)_2$ in acetic acid.
- **Basic beryllium nitrate [$Be_4O(NO_3)_6$]** shares the same Be_4O core; NO_3^- instead bridges pairs of Be atoms bidentately.

8 · STRUCTURE — CAC_2 LATTICE & THE CYANAMIDE ION



Distorted NaCl-type: Ca^{2+} ↔ Na^+, linear C_2^{2-} ↔ Cl^- (non-spherical → distortion; >450 °C → tetragonal, disordered)

$CaC_2 + N_2 \xrightarrow{(1100°C)} CaNCN + C$ — industrial N_2 fixation

$[N{=}C{=}N]^{2-}$, linear — isoelectronic with CO_2

$CaNCN + 5H_2O \rightarrow CaCO_3 + 2NH_4OH$ (slow-release fertilizer) · → urea, thiourea, melamine downstream

PART B — COMPLETE NOTES (FULL TEXT, NCERT + ALLEN + J.D. LEE/GUHA)

▮ ABOUT THESE NOTES

JEE ADVANCED

MASTER NOTES

ALKALINE EARTH METALS

Group 2 · s-Block Elements

Be · Mg · Ca · Sr · Ba · Ra

NCERT FOCUS — How to read these notes

Built from NCERT (Unit 10, Class XI Chemistry), J.D. Lee's *Concise Inorganic Chemistry* (Group 2), and JEE-Advanced coaching sources, cross-checked for consistency. Green boxes mark NCERT-exact scope/definitions/uses; blue boxes give molecular-orbital / valence-bond reasoning for bonding; orange boxes flag traps examiners exploit.

Compiled for rigorous, theory-first, reaction-complete revision.

▮ 1. INTRODUCTION & GENERAL TRENDS

Group 2 comprises **Be, Mg, Ca, Sr, Ba, Ra** — general valence configuration *ns2*. All members except Be are traditionally called **alkaline earth metals** because their oxides/hydroxides are alkaline and the metals occur combined as "earths" (refractory oxide minerals) in nature.

1.1 Electronic Configuration & Oxidation State

Element	Symbol	Configuration	Atomic No.
Beryllium	Be	[He] $2s^2$	4
Magnesium	Mg	[Ne] $3s^2$	12
Calcium	Ca	[Ar] $4s^2$	20
Strontium	Sr	[Kr] $5s^2$	38
Barium	Ba	[Xe] $6s^2$	56
Radium	Ra	[Rn] $7s^2$	88

+2 is the only significant oxidation state. Removing both *ns* electrons leaves a noble-gas core, so the **third ionization enthalpy is enormous** (e.g. IE_3 of Be ≈ 14,847 kJ/mol, Mg ≈ 7,731 kJ/mol) — M^{3+} is never formed.

1.2 Occurrence (minerals worth remembering)

- **Be:** beryl $Be_3Al_2Si_6O_{18}$ (rare)
- **Mg:** magnesite $MgCO_3$, dolomite $CaCO_3 \cdot MgCO_3$, carnallite $KCl \cdot MgCl_2 \cdot 6H_2O$, seawater
- **Ca:** limestone/chalk/marble ($CaCO_3$), gypsum $CaSO_4 \cdot 2H_2O$, fluorite CaF_2, fluorapatite $3Ca_3(PO_4)_2 \cdot CaF_2$
- **Sr:** celestite $SrSO_4$, strontianite $SrCO_3$
- **Ba:** barytes $BaSO_4$, witherite $BaCO_3$
- **Ra:** extreme trace, radioactive; isolated by the Curies from pitchblende

1.3 Atomic & Ionic Radii

Radii increase steadily down the group as a new shell is added each period. At the same time, Group 2 atoms/ions are smaller than the corresponding Group 1 element of the same period, because the extra proton (higher nuclear charge) pulls the electron cloud in more tightly.

Quantity	Be	Mg	Ca	Sr	Ba
Metallic radius (pm)	111	160	197	215	222
M^{2+} ionic radius (pm)	31	72	100	118	135

1.4 Ionization Enthalpy

	Be	Mg	Ca	Sr	Ba	Ra
IE_1 (kJ/mol)	899	737	590	549	503	509
IE_2 (kJ/mol)	1757	1450	1145	1064	965	979

Ra's IE_1/IE_2 are very slightly **higher** than Ba's — this is not a random anomaly. Radium comes right after the **$4f^{14}$ subshell** is filled (through the lanthanides) and after the 5d subshell too; the poorly-shielding, diffuse 4f electrons let the nuclear charge "leak through" more effectively to the outer 7s electrons — the same *lanthanide-contraction-type* effect responsible for Cs vs. Fr and other 6th/7th-period irregularities. So Ra is very slightly **smaller and harder to ionize** than a naive extrapolation down the group would predict.

> **NCERT FOCUS — Why is IE_1(Group 2) > IE_1(Group 1) but IE_2(Group 2) < IE_2(Group 1) for the same period?**
>
> Group 2 atoms are smaller than the Group 1 atom of the same period (higher nuclear charge), so their **IE_1 is higher**. But a Group 1 atom, having lost one electron, already has a noble-gas core — pulling out a **second** electron from that core (IE_2 of Group 1) is extremely costly. For Group 2, the second electron removed is the one that *completes* the noble-gas configuration, so **IE_2 of Group 2 < IE_2 of Group 1** for the same period.

1.5 Hydration Enthalpy

M^{2+}	Be^{2+}	Mg^{2+}	Ca^{2+}	Sr^{2+}	Ba^{2+}
ΔH_hyd (kJ/mol)	−2494	−1921	−1577	−1443	−1305

Hydration enthalpies fall steadily as ionic size increases (weaker ion–dipole pull). Group 2 values run **4–5× larger** than the corresponding Group 1 ion (smaller size, double the charge). This is why Group 2 salts crystallize with far more water of hydration than Group 1 salts: *$MgCl_2{\cdot}6H_2O$, $CaCl_2{\cdot}6H_2O$, $SrCl_2{\cdot}6H_2O$, $BaCl_2{\cdot}2H_2O$* exist, while NaCl and KCl are anhydrous. Note the number of hydrate waters **falls** down the group as hydration enthalpy weakens.

1.6 Melting Point, Boiling Point, Density

	Be	Mg	Ca	Sr	Ba	Ra
m.p. (K)	1560	924	1124	1062	1002	973
b.p. (K)	2745	1363	1767	1655	2078	≈1973
Density (g/cm^3)	1.84	1.74	1.55	2.63	3.59	≈5.5

The actual crystal structures behind the "different lattice" explanation: Be and Mg adopt **hexagonal close-packed (hcp, CN = 12)**; Ca and Sr adopt **cubic close-packed (ccp/fcc, CN = 12)** at room temperature; Ba adopts **body-centred cubic (bcc, CN = 8)**. Ca even undergoes its own *fcc → bcc phase transition* on heating. Since bcc packing is less efficient (lower coordination number, lower packing fraction) than hcp/ccp, the metallic bonding is weaker wherever bcc appears — this is precisely why Ba (bcc) breaks the naive "m.p. falls smoothly down the group" pattern.

> **JEE TRAP**
>
> - The m.p./b.p./density trend down the group is **NOT monotonic** — do not force a simple "decreases smoothly" or "increases smoothly" narrative. NCERT itself flags this explicitly: the metals crystallize in **different lattice structures**, so the trend is irregular.
> - Group 2 metals are harder and higher-melting than the corresponding Group 1 metals because

two valence electrons (not one) participate in metallic bonding — roughly double the cohesive energy.

1.7 Standard Reduction Potential & Reducing Power

M^{2+}/M	Be	Mg	Ca	Sr	Ba	Ra
E° (V), J.D. Lee Table 11.9	−1.85	−2.37	−2.87	−2.89	−2.91	−2.92

- Reducing power broadly increases Be → Ba, but **Be is anomalous** — its E° is **least negative** in the group (weakest reducing agent). This indicates Be is much less electropositive (less "metallic") than the rest of the group.
- Ca, Sr, Ba have reduction potentials **similar to the corresponding Group 1 metals**, and are quite high in the electrochemical series.

NCERT FOCUS — NCERT's exact reasoning for Be's weak reducing power (matches J.D. Lee's Born–Haber-type logic)

"Beryllium has less negative value compared to other alkaline earth metals. However, its reducing nature is due to large hydration energy associated with the small size of Be^{2+} ion and relatively large value of the atomization enthalpy of the metal."

Key idea: E° is a **net thermodynamic sum** of atomization enthalpy + IE_1 + IE_2 + hydration enthalpy — it cannot be read off from ionization enthalpy alone. This mirrors the classic Li-anomaly in Group 1.

JEE TRAP

- Different textbooks tabulate E°(Be^{2+}/Be) as either −1.85 V (J.D. Lee, and used in Be–Al comparisons) or −1.97 V (some standard data tables) — both are legitimate literature values (method-dependent). Use **−1.85 V** as the primary JEE-quotable figure since it is the one that correctly places Be closest to Al (−1.66 V) among all metals.

1.8 Flame Colours

Metal	Ca	Sr	Ba	Be, Mg
Flame colour	Brick red	Crimson	Apple green	None

Mechanism: flame heat promotes an outer electron to a higher level; on relaxation the energy is emitted as visible light ($E = h\nu$). **Be and Mg show no flame colour** because their valence electrons are too strongly (tightly) bound by the small, high-charge-density nucleus to be excited at ordinary flame temperatures.

JEE TRAP

- "Mg burns with a dazzling white light" describes **combustion emission** (incandescent MgO), *not* a flame-test colour for Mg^{2+}. Mg still gives **no** characteristic flame-test line.

1.9 Extraction of the Metals (J.D. Lee)

Group 2 metals are themselves strong reducing agents and are strongly electropositive, so aqueous solutions cannot be used to displace them with another metal, and simple chemical reduction of the oxides is not viable. Electrolysis of aqueous solutions is possible using a mercury cathode (giving an amalgam), but recovering the pure metal from the amalgam is difficult. **All the metals can be obtained by electrolysis of the fused chloride**, with NaCl added to lower the melting point, though Sr and Ba tend to form a colloidal suspension during the process.

Beryllium

- Extracted from beryl $Be_3Al_2Si_6O_{18}$ by fusion or sulphuric acid treatment to give soluble $BeSO_4$, which is treated with NH_4OH to precipitate the hydroxide $Be(OH)_2$ (BeO is amphoteric, but the aluminium is removed as the soluble tetrahydroxoaluminate complex).
- $Be(OH)_2$ is converted to BeF_2 (by treatment with HF), and the metal is obtained either by **reduction with Mg**: $BeF_2 + Mg \rightarrow Be + MgF_2$, or by electrolysis of the fused chloride made from BeO + C + Cl_2.
- Both Be and BeO must be handled with extreme purity control — the main use of the metal is as a neutron moderator/reflector in nuclear reactors, and for windows on X-ray tubes because of its very low electron density (low X-ray absorption).

Magnesium

$$MgO + C \xrightarrow{2000\ °C} Mg + CO$$

(quenched rapidly — the reverse reaction is thermodynamically favoured on cooling)

- **Pidgeon process:** calcined dolomite [$CaCO_3 \cdot MgCO_3 \rightarrow CaO \cdot MgO$] is reduced by heating with **ferrosilicon** at 1150 °C under reduced pressure:

$$[CaCO_3 \cdot MgCO_3] \xrightarrow{heat} CaO \cdot MgO \xrightarrow{+Fe/Si} Mg + Ca_2SiO_4 + Fe$$

- **Dow sea water process:** seawater (≈0.13% Mg^{2+}) is treated with slaked lime — $Mg(OH)_2$ is far less soluble than $Ca(OH)_2$, so it precipitates and is filtered off, then converted to $MgCl_2$ with HCl and electrolysed:

$$Ca(OH)_2 + MgCl_2(seawater) \rightarrow Mg(OH)_2\downarrow + CaCl_2$$
$$Mg(OH)_2 + HCl \xrightarrow{heat} MgCl_2$$

- **Dow natural brine process:** calcined dolomite ($MgO \cdot CaO$) is treated with HCl to give a $CaCl_2/MgCl_2$ solution; adding more calcined dolomite plus CO_2 precipitates $CaCO_3$, leaving pure $MgCl_2$ solution for electrolysis:

$$CaCl_2 \cdot MgCl_2 + CaO \cdot MgO + 2CO_2 \rightarrow 2MgCl_2 + 2CaCO_3\downarrow$$

Calcium, Strontium, Barium

Ca metal is obtained by electrolysis of fused $CaCl_2$ (from the Solvay-process waste stream, or from $CaCO_3$ + HCl). World production of Ca metal is only about 1000 tonnes/year; it is used in Al alloys, in the iron/steel industry to control carbon and scavenge P, O and S, as a reducing agent for Zr/Cr/Th/U, and to remove traces of N_2 from argon. Sr and Ba are produced on a much smaller scale, by electrolysis of the fused chloride, or by thermite-type reduction of the oxide with aluminium.

1.10 Uses of the Group 2 Metals Themselves (NCERT §10.6.7)

NCERT FOCUS — NCERT-exact uses list — the metals, not their compounds

- Beryllium: used in the manufacture of alloys — copper–beryllium alloys are used to prepare high-strength springs. Metallic beryllium is used for making windows of X-ray tubes.
- Magnesium: forms alloys with aluminium, zinc, manganese and tin. Mg–Al alloys, being light in mass, are used in aircraft construction. Magnesium (powder and ribbon) is used in flash powders and bulbs, incendiary bombs, and signals. A suspension of magnesium hydroxide in water ("milk of magnesia") is used as an antacid in medicine. Magnesium carbonate is an ingredient of toothpaste.
- Calcium: used in the extraction of metals from oxides which are difficult to reduce with carbon. Calcium and barium metals, owing to their reactivity with oxygen and nitrogen at elevated temperatures, have often been used to remove the last traces of air from vacuum tubes (as "getters").
- Radium: radium salts are used in radiotherapy, for example in the treatment of cancer.

1.11 Lattice Energy vs Hydration Energy — The Master Solubility Framework

Solubility of Group 2 compounds departs from the usual "heavier = less soluble" intuition and is instead governed entirely by the **competition between lattice energy and hydration energy** as the cation grows down the group.

Lattice energy (kJ/mol)	Mg	Ca	Sr	Ba
MO	−3923	−3517	−3312	−3120
MCO_3	−3078	−2986	−2718	−2614
MF_2	−2906	−2610	−2496	−2367
MCl_2	−2592	−2258	−2159	−2018 (approx.)

ΔH_hydration	Be^{2+}	Mg^{2+}	Ca^{2+}	Sr^{2+}	Ba^{2+}

(kJ/mol)					
	−2494	−1921	−1577	−1443	−1305

MOT / BONDING — J.D. Lee's exact reasoning (the single most important paragraph in the chapter)

"For a substance to dissolve, the hydration energy must exceed the lattice energy. Consider a related group of compounds, such as the chlorides of all the Group 2 metals. On descending the group, the metal ions become larger and so both the lattice energy and the hydration energy decrease. A decrease in lattice energy favours increased solubility, but a decrease in hydration energy favours decreased solubility. These two factors thus change in opposite directions, and the overall effect depends on which of the two has changed most. With most compounds, on descending the group, the hydration energy decreases more rapidly than the lattice energy: hence the compounds become less soluble as the metal gets larger. However, with fluorides and hydroxides the lattice energy decreases more rapidly than the hydration energy, and so their solubility increases on descending the group."

▮ 2. CHEMICAL REACTIVITY

2.1 Reactivity toward Air / O_2 / N_2

- **Be, Mg:** kinetically inert at room temperature — a thin, adherent protective oxide film forms and stops further attack. The metals are relatively unreactive in massive form and do not react below about 600 °C, but powdered metal is far more reactive: powdered Be ignites to give BeO + Be_3N_2. Mg burns with dazzling brilliance, evolving a great deal of heat — this is used to **start a thermite reaction** with aluminium, and to provide light in (old-style) flash photography using light bulbs, not electronics, giving MgO + Mg_3N_2.
- **Ca, Sr, Ba:** tarnish readily even at ordinary temperature, forming oxide + nitride.
- Normal oxide MO forms on burning in excess O_2 for all members; **Ba additionally forms the peroxide BaO_2** ($2Ba + O_2 \rightarrow 2BaO_2$) — peroxide-forming tendency **increases with cation size** (a larger, less polarizing cation stabilizes the large, diffuse O_2^{2-} ion better).

JEE TRAP

- Burning Mg is **not** extinguished by CO_2 — Mg is a strong enough reductant to reduce it: $2Mg + CO_2 \rightarrow 2MgO + C$. CO_2 extinguishers must never be used on a magnesium fire.

2.2 Reactivity toward Water

Metal	Behaviour
Be	Essentially inert; protected by BeO film (sources note some doubt whether it reacts with steam at all)
Mg	No reaction with cold water; slowly decomposes hot water → $Mg(OH)_2$ + H_2; with steam → MgO + H_2
Ca, Sr, Ba	React readily with cold water: $M + 2H_2O \rightarrow M(OH)_2 + H_2\uparrow$

Magnesium's E° (−2.37 V) is actually favourable for reaction, but Mg **forms a protective oxide layer** — so despite this favourable reduction potential it does not readily react with cold water unless the oxide layer is first removed, e.g. by **amalgamating the surface with mercury**. In this particular respect (a favourable driving force blocked by a passivating oxide film), Mg's behaviour resembles that of aluminium.

> **JEE TRAP**
>
> - The isolated solid product from Mg + water depends on **conditions**: *hot liquid water* → $Mg(OH)_2 + H_2$, but *steam* → $MgO + H_2$ (dehydration favoured at higher T). "Mg + steam → $Mg(OH)_2$" is a commonly-set wrong option.
> - Be doesn't react with water because of **kinetic passivation**, not because it's thermodynamically unreactive — same logic pattern as Li in Group 1 (most negative E°, yet reacts gently with water due to a high melting point/low surface-area factor). Never infer reactivity from E° alone.

2.3 Reactivity toward Dihydrogen — Hydrides

Ca, Sr, Ba (and Mg, with more difficulty) react directly with H_2 on heating to give MH_2: $Ca + H_2 \rightarrow CaH_2$. BeH_2 cannot be made this way and must instead be made **indirectly**, by two independent routes:

$$2BeCl_2 + LiAlH_4 \rightarrow 2BeH_2 + LiCl + AlCl_3$$

Mechanistic picture: $LiAlH_4$ is a source of the hydride nucleophile H^-. In effect, H^- attacks the small, electron-deficient, highly electrophilic Be centre and displaces Cl^- — an **SN2-like nucleophilic substitution** at beryllium, repeated until both chlorines are replaced by hydride.

A purer sample (free of Al/Li contamination) is obtained via a two-step borohydride route:

$$BeCl_2 + 2Li[BH_4] \rightarrow BeB_2H_8 + 2LiCl$$
$$BeB_2H_8 + 2PPh_3 \xrightarrow{\text{(sealed tube, heat)}} BeH_2 + 2Ph_3PBH_3$$

Ionic vs covalent nature: CaH_2, SrH_2, BaH_2 are **ionic** (contain discrete H^-); BeH_2 and MgH_2 are **covalent and polymeric**. All hydrides are strong reducing agents and hydrolyse in water/dilute acid liberating H_2:

$$CaH_2 + 2H_2O \rightarrow Ca(OH)_2 + 2H_2$$
$$(CaH_2 = \text{"hydrolith"})$$

> **MOT / BONDING — Why is BeH_2 electron-deficient & polymeric?**
>
> Ground-state Be is $1s^2 2s^2$ — **no unpaired electrons**, so it cannot form two ordinary covalent bonds directly. Promoting one 2s electron to 2p gives $1s^2 2s^1 2p^1$ (two unpaired electrons, **sp hybridisation**) — this lets an *isolated gas-phase* BeH_2/BeX_2 molecule be **linear** (180°).
>
> But even after sp-hybridised bonding, Be's outer shell holds only 4 electrons around it — it is **electron-deficient**. In the condensed (solid) phase Be resolves this by forming **3-centre–2-electron ("banana") bonds**: each Be–H–Be bridge is one shared electron pair spread over three atoms, so the polymer chain gives every Be a coordination number of 4 despite Be starting with only 2 valence electrons. This is the same *electron-deficient cluster bonding* logic used for diborane B_2H_6.

2.4 Reactivity toward Halogens — Halides

$M + X_2 \rightarrow MX_2$ for all members. **Except beryllium halides, all Group 2 halides are ionic.** $BeCl_2$ and BeF_2 are prepared as:

$$BeO + C + Cl_2 \xrightarrow{\text{(600–800 K)}} BeCl_2 + CO$$
$$2BeO + CCl_4 \xrightarrow{\text{(800 °C)}} 2BeCl_2 + CO_2$$

$$BeO + 2NH_3 + 4HF \rightarrow (NH_4)_2BeF_4 \xrightarrow{\text{(heat)}} BeF_2 + 2NH_4F$$
$$[(NH_4)_2BeF_4 = \text{ammonium fluoroberyllate}]$$

Anhydrous Be/Mg halides **cannot** be obtained by simply heating the hydrate — this causes hydrolysis instead of dehydration:

$$[Be(H_2O)_4]Cl_2 \xrightarrow{\text{(heat)}} Be(OH)_2 + 2HCl + 2H_2O$$
$$MgCl_2 \cdot 6H_2O$$

```
→(heat)→
MgO + 2HCl + 5H₂O
```

This is a general rule for **small, highly-charged, strongly polarising cations** — the same hydrolysis-on-heating problem is seen with Be^{2+}, Mg^{2+}, Al^{3+}, Cr^{3+}, Fe^{3+} and similar high-charge-density ions across the periodic table, not just Group 2. Contrast this with Ca, Sr, Ba, where dehydration proceeds cleanly on heating with no hydrolysis:

```
CaCl₂·6H₂O
→(heat)→
CaCl₂ + 6H₂O
(simple dehydration — no hydrolysis,
because Ca²⁺ is a much weaker polariser)
```

(Industrially, anhydrous $MgCl_2$ is instead made by heating the double salt $MgCl_2·NH_4Cl·6H_2O$, which loses water first and NH_4Cl only at higher T, avoiding hydrolysis:

```
MgCl₂·NH₄Cl·6H₂O
→(−H₂O, Δ)→
MgCl₂·NH₄Cl
→(strong Δ)→
MgCl₂ + NH₃ + HCl
```

Anhydrous $CaCl_2$ is an excellent drying agent for gases and organic vapours — **but not for NH_3 or ethanol**, since it forms stable adducts instead: $CaCl_2·8NH_3$ and $CaCl_2·4C_2H_5OH$. A JEE-favourite exception to "$CaCl_2$ dries everything."

MOT / BONDING — $BeCl_2$ structure — solid, dimer, and monomer

Solid state: an infinite zig-zag **chain polymer** with chlorine bridges — each Cl makes one normal covalent bond to one Be and donates a lone pair (dative bond) to the next Be, giving every Be a coordination number of 4.

Vapour, moderate T: a chloro-bridged **dimer** Be_2Cl_4.

Vapour, ~1200 K: dissociates into the **linear monomer** (sp-hybridised, 2-coordinate, electron-deficient).

Contrast: in $Be(CH_3)_2$ the bridge is a genuine **3-centre-2-electron bond** spanning Be···CH_3···Be (like BeH_2), *not* simple lone-pair donation as in the chlorine bridges of $(BeCl_2)_n$ — a subtle but real mechanistic difference between halide bridging (dative) and alkyl bridging (electron-deficient).

Covalency trend (Fajans' rules)

Covalent character: **BeX_2 > MgX_2 > CaX_2 > SrX_2 > BaX_2** — the small, highly-charged Be^{2+} polarizes the anion's electron cloud strongly ("small highly charged ions tend to form covalent compounds"). For a fixed cation, covalency rises with anion polarizability: I^- > Br^- > Cl^- > F^-.

JEE TRAP

- BeF_2 has the **largest** electronegativity difference of any Be halide, yet is still regarded as covalent when fused (very low melt conductivity) — high charge density beats electronegativity difference for Be specifically.
- BeF_2 is, unusually, **very soluble** in water (owing to the huge solvation energy that forms $[Be(H_2O)_4]^{2+}$), even though heavier Group-2 fluorides are all almost insoluble — MF_2 (M = Mg, Ca, Sr, Ba) are white, insoluble, high-melting ionic solids.
- Hydration of halides **decreases** down the group: $CaCl_2·6H_2O$, $SrCl_2·6H_2O$, but **$BaCl_2·2H_2O$** only. For $MgCl_2$, NCERT itself is internally inconsistent — §10.6.4 states $MgCl_2·6H_2O$, but the halides table in §10.7 states $MgCl_2·8H_2O$ for the very same salt. Both forms appear in circulation; if a question is drawn straight from NCERT's halides section, expect **$8H_2O$**; otherwise **$6H_2O$** is the more common literature value (matching J.D. Lee).

Chlorides, bromides, iodides — solubility and industrial uses

The chlorides, bromides and iodides of Mg, Ca, Sr, Ba are ionic, have much lower melting points than the fluorides, and are readily soluble in water; solubility falls off somewhat with increasing atomic number. All these halides form hydrates and are **hygroscopic** (absorb water vapour from the air). **CaF_2** is the main industrial source of both F_2 and HF:

```
CaF₂ + H₂SO₄ → 2HF + CaSO₄
HF + KF
```

$$\xrightarrow{\text{(electrolysis)}} F_2$$

Several million tonnes of **$CaCl_2$** are produced annually, mostly discarded as a low-value Solvay-process by-product. Uses: treating ice on roads (a 30% $CaCl_2/H_2O$ eutectic freezes at −55 °C, vs −18 °C for $NaCl/H_2O$ — far more effective in very cold climates); making concrete set faster and stronger; as a laboratory desiccant (drying agent) owing to its strong hygroscopicity; and (via the electrolytic route) as the feedstock for extracting Ca metal.

2.5 Reactivity toward Acids and Alkalis

$$M + 2HCl \rightarrow MCl_2 + H_2\uparrow \quad \text{(Mg, Ca, Sr, Ba)}$$

Be reacts only after its oxide film is removed, and — like Al — is **rendered passive by concentrated HNO_3** (a thin protective oxide layer forms). **Be is amphoteric even as the metal**, dissolving in NaOH:

$$Be + 2NaOH + 2H_2O \rightarrow Na_2[Be(OH)_4] + H_2\uparrow \quad \text{(sodium beryllate)}$$

Mg, Ca, Sr, Ba do **not** react with NaOH — they are purely basic metals.

3. REDUCING NATURE & SOLUTIONS IN LIQUID AMMONIA

Ca, Sr, Ba dissolve in liquid NH_3 to give **deep blue** solutions — identical phenomenon to Group 1 — due to **ammoniated (solvated) electrons**:

$$M + (x+y)NH_3 \rightarrow [M(NH_3)_x]^{2+} + 2[e(NH_3)_y]^-$$

- Concentrated solutions turn **bronze-coloured** (metal clusters form).
- On standing/evaporation, Group 2 solutions do **not** simply deposit the metal back (unlike Group 1) — evaporation instead gives crystalline **hexammoniates** $M(NH_3)_6$, which slowly decompose to the amide:

$$M(NH_3)_6 \rightarrow M(NH_2)_2 + 4NH_3 + H_2\uparrow$$

Mg is far less soluble in liquid NH_3 under ordinary conditions; Be does not typically show this behaviour. Net reducing action of the solution: $2NH_3 + 2e^- \rightarrow 2NH_2^- + H_2$ — the ammoniated electron acts as a powerful reducing agent / very strong base.

▮ 4. NITRIDES AND CARBIDES

4.1 Nitrides

All Group 2 metals burn in N_2 to form **ionic nitrides M_3N_2** — in sharp contrast to Group 1, where only Li (the smallest, most polarizing) forms a nitride Li_3N. The reason is the same in both groups: nitride formation needs a very large **lattice energy** to overpay for breaking the strong N≡N bond, and lattice energy is large whenever both ions carry high charge (M^{2+}, N^{3-}) — so it works for the *whole* of Group 2, not just the smallest member.

$$Ca_3N_2 + 6H_2O \rightarrow 3Ca(OH)_2 + 2NH_3\uparrow$$
(all nitrides hydrolyse this way)

Be_3N_2 is comparatively volatile (greater covalent character of Be); the other nitrides are not.

4.2 Carbides

Carbide	Anion	Hydrolysis product	Note
Be_2C	C^{4-} ("methanide")	CH_4	Antifluorite structure; $Be_2C + 4H_2O \rightarrow 2Be(OH)_2 + CH_4$
Mg_2C_3	C_3^{4-} ("allylide")	$CH_3-C\equiv CH$ (propyne)	Not on the acetylide pattern
CaC_2, SrC_2, BaC_2	C_2^{2-} ("acetylide")	C_2H_2 (ethyne)	Distorted NaCl-type lattice (C_2^{2-} is non-spherical)

Be_2C is made by heating BeO with carbon at 1900–2000 °C; it is brick-red and adopts an **antifluorite structure** (C^{4-} replaces F^-, Be^{2+} replaces Ca^{2+}).

$$CaC_2 + 2H_2O \rightarrow Ca(OH)_2 + C_2H_2\uparrow$$
$$CaO + 3C \xrightarrow{(2000°C)} CaC_2 + CO$$
$$Ca + 2C \xrightarrow{(1100°C)} CaC_2$$

The MC_2 carbides (Ca, Sr, Ba) all adopt a **distorted sodium chloride-type structure**: M^{2+} replaces Na^+ and the linear $C\equiv C^{2-}$ ion replaces Cl^-, but because C_2^{2-} is not spherical (unlike Cl^-) the lattice is distorted along the axis where the ions are aligned, and the unit cell is tetragonal at room temperature. **Above 450 °C** the C_2^{2-} ions adopt random orientations rather than staying aligned, and the cell becomes genuinely tetragonal in a different, higher-symmetry sense.

At one time the main source of ethyne (acetylene) for oxy-acetylene welding was CaC_2; world production peaked at 7 million tonnes/year in 1960, but had declined to 4.9 million tonnes by 1991 as ethyne became more cheaply obtained from processing oil.

JEE TRAP

- Not every Group-2 carbide gives ethyne! Only the true **acetylides** (Ca, Sr, Ba) do. Be_2C gives methane; Mg_2C_3 gives propyne.

4.3 Calcium Cyanamide — Full Reaction Network

CaC_2 is also an important industrial intermediate: heated in an electric furnace with N_2 at 1100 °C, it gives **calcium cyanamide, CaNCN** — an important route of fixing atmospheric nitrogen (an alternative to the Haber process).

$$CaC_2 + N_2 \xrightarrow{(1100\ °C,\ \text{electric furnace})} CaNCN + C$$

The cyanamide ion $[N=C=N]^{2-}$ is **isoelectronic with CO_2** and linear. CaNCN is produced on a large scale (particularly in locations where there is cheap electricity) and used directly, on a huge scale, as a **slow-acting nitrogenous fertilizer** (widely used in SE Asia and the Far East) — it hydrolyses slowly over months, giving it a genuine advantage over more soluble fertilizers like NH_4NO_3 or urea, which are washed away by the first rainstorm:

$$CaNCN + 5H_2O \rightarrow CaCO_3 + 2NH_4OH$$

CaNCN is also the starting material for several other important industrial products:

$$CaNCN + H_2SO_4 \rightarrow H_2NCN + CaSO_4$$
$$CaNCN + CO_2 + H_2O \rightarrow H_2NCN + CaCO_3$$
(cyanamide, H_2NCN)

$$H_2NCN + H_2O \xrightarrow{(pH < 2 \text{ or } > 12)} H_2N{\cdot}CO{\cdot}NH_2$$
(urea)

$$H_2NCN + H_2S \rightarrow H_2N{\cdot}CS{\cdot}NH_2$$
(thiourea)

$$H_2NCN \xrightarrow{(pH\ 7\text{–}9)} NCNC(NH_2)_2 \text{ (dicyanamide)} \xrightarrow{(pyrolysis)} \text{cyanuric amide (melamine)}$$

Melamine — the ring compound formed by pyrolysing dicyanamide — reacts with formaldehyde to form hard plastics (melamine resins), so this entire chain traces a path from N_2 gas all the way to a common household plastic.

JEE TRAP

- $CaC_2 + N_2$ (1100 °C, electric furnace) gives **calcium cyanamide CaNCN** + C, *but* **BaC_2 + N_2** gives a **cyanide** $Ba(CN)_2$, *not* a cyanamide — a genuinely exception-worthy fact worth memorising as a pair.
- Do not confuse the *three* nitrogen-containing products in this chain: **cyanamide** (H_2NCN, the free acid), **dicyanamide** ($NCNC(NH_2)_2$, its pH 7–9 dimer), and **melamine** (the pyrolysed cyclic trimer-type product) — each forms under a different, specific condition from the last.

5. COMPLEX-FORMING TENDENCY

Group 2 ions form complexes more readily than Group 1 (smaller size → higher charge density), but far less readily than the transition metals. **Be**, being appreciably smaller than the rest, is by far the best complex-former of the group; only Mg and Ca show any real tendency to complex, mostly with O-donor ligands.

- **$[BeF_4]^{2-}$** (tetrafluoroberyllate) — Be readily coordinates two extra F^- ions; the $M_2[BeF_4]$ salts closely resemble sulphates in their solubility properties, correlating with the fact that Be, uniquely in the group, has empty orbitals available for complex formation.
- **$BeCl_2{\cdot}D_2$ adducts** (D = an ether, aldehyde or ketone with an O lone pair) — tetrahedral, formed the same way as $[BeF_4]^{2-}$.
- **$[Be(H_2O)_4]^{2+}$, $[Be(OH)_4]^{2-}$** — always **tetrahedral**, never 6-coordinate, because Be has only 4 valence orbitals (one 2s + three 2p) available for bonding, with no accessible d-orbitals.
- Stable chelates: **beryllium oxalate $[Be(ox)_2]^{2-}$**, and complexes with β-diketones (acetylacetone) and catechol — all tetrahedral at Be.
- **Basic beryllium acetate $[Be_4O(CH_3COO)_6]$**: a central O^{2-} surrounded tetrahedrally by 4 Be atoms, with 6 acetate groups bridging the 6 tetrahedron edges. Covalent, low-melting (285 °C)/b.p. 330 °C, soluble in organic solvents, and can even be distilled — this volatility/solubility is exploited to **purify beryllium**. Prepared by evaporating $Be(OH)_2$ in acetic acid. A near-identical structure is adopted by **basic beryllium nitrate $[Be_4O(NO_3)_6]$**, where each NO_3^- instead bridges a pair of Be atoms bidentately.
- Ca^{2+}/Mg^{2+} form a stable 1:1 chelate with **EDTA** (hexadentate) — $[Ca(EDTA)]^{2-}$ is **six-coordinate** (octahedral), containing **5 five-membered chelate rings**, the basis of complexometric titration for water-hardness estimation. Interestingly, Be — despite being the strongest complexer in the group — does **not** complex appreciably with EDTA, because EDTA needs 6 coordination sites and Be is invariably 4-coordinate. Ca^{2+}/Mg^{2+}-EDTA titrations are performed at a **higher pH** than titrations of other metals (e.g. Zn^{2+}, Cd^{2+}, Pb^{2+}), because the Ca/Mg-

EDTA complexes are less stable, and at low pH EDTA gets protonated instead of complexing.

- **Chlorophyll**: Mg^{2+} sits at the centre of a flat porphyrin ring, bonded to 4 N atoms; the complex absorbs light in the red region and drives photosynthesis, $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$. Mg forms a few halide complexes too, such as $[NEt_4]_2[MgCl_4]$, but Ca, Sr, Ba do not form comparable halide complexes.

JEE TRAP

- Be salts taste sweet but must *never* be tested by taste — Be compounds are highly toxic (they displace Mg^{2+} from enzymes; inhaling Be dust or smoke causes berylliosis, a disease resembling silicosis; skin contact causes dermatitis).

6. ORGANOMETALLIC COMPOUNDS — GRIGNARD REAGENTS & BE/MG ALKYLS

Both Be and Mg form an appreciable number of M–C bonded compounds; only a few have been isolated for Ca, Sr, Ba. **Grignard reagents** (organomagnesium halides, RMgX) are probably the single most versatile class of reagent in organic chemistry, and — along with lithium alkyls — provide the two general routes to organometallic compounds across the periodic table. Victor Grignard won the 1912 Nobel Prize in Chemistry for this work.

6.1 Preparation

```
Mg + RBr
→(dry ether)→
RMgBr
(Grignard reagent)
```

- Made by the slow addition of an alkyl or aryl halide (Cl, Br or I) to a continuously stirred mixture of magnesium turnings in an absolutely dry organic solvent, usually diethyl ether. The reaction is often slow to start and may require an induction period before it starts — addition of a crystal of iodine, or gently warming, is used to penetrate the metal's oxide film and initiate the reaction.
- Reactivity order: **iodides** are the most reactive, **chlorides** the least reactive; alkyl compounds usually form Grignard reagents more readily than aryl compounds.
- Grignard reagents are rapidly hydrolysed by water to give the parent hydrocarbon, so they are never isolated solid — made and used *in situ*:

$$2RMgBr + 2H_2O \rightarrow 2RH + Mg(OH)_2 + MgBr_2$$

6.2 Structure

Grignard reagents are not simple — they are normally solvated/polymerized without halogen bridges. Three-centre structures have long been the subject of controversy; X-ray structures of solid $PhMgBr{\cdot}2Et_2O$ and $EtMgBr{\cdot}2Et_2O$ show magnesium is **tetrahedrally coordinated** by the organic group, oxygen from ether molecules, and (in solution) several other species may be present.

6.3 Reactions of Grignard Reagents (complete synthetic-use list)

Reagent added to RMgBr	Product (after acid/H_2O work-up)
CO_2	R·COOH (carboxylic acid)
$R'_2C{=}O$	$R'_2C(OH)R$ (tertiary alcohol)
R'·CHO	R'CH(OH)R (secondary alcohol)
HCHO	$R{\cdot}CH_2OH$ (primary alcohol)
O_2	R·OH
S_8	RSH and R_2S
I_2	RI
H^+	RH

Grignard reagents also transfer their organic group to other element halides, providing routes to a wide range of organometallic and organic products (R = alkyl or aryl):

$$RMgBr + BeCl_2 \rightarrow BeR_2$$
$$RMgBr + LiR \rightarrow MgR_2$$
$$RMgBr + BCl_3 \rightarrow BR_3$$

$$RMgBr + SiCl_4 \rightarrow RSiCl_3,\ R_2SiCl_2,\ R_3SiCl,\ R_4Si$$
(alkyl/aryl chlorosilanes — commercially important in making silicones)

6.4 Beryllium and Magnesium Dialkyls/Diaryls

$BeCl_2$ reacts with Grignard reagents to give reactive alkyls and aryls:

$$BeCl_2 + 2MeMgCl \xrightarrow{(Et_2O)} BeMe_2{\cdot}(Et_2O)_n + 2MgCl_2$$
$$BeCl_2 + 2LiEt \xrightarrow{(Et_2O)} BeEt_2{\cdot}(Et_2O)_n$$
$$Be + HgMe_2 \xrightarrow{(warm)} BeMe_2 + Hg$$

$BeMe_2$ is dimeric in the vapour but **polymerized in the solid**, with a chain structure resembling that of $BeCl_2$ — but the bonding is different: the Be–Me–Be bridge is a genuine **3-centre-2-electron bond** (like BeH_2), unlike the lone-pair-donor halogen bridges of $(BeCl_2)_n$. Similar dialkyl/diaryl compounds of Ca, Sr, Ba can be made using Grignard reagents, lithium alkyls/aryls, or mercury alkyls/aryls — the **Ca, Sr, Ba compounds are much more reactive than the corresponding Mg compound**. The beryllium alkyls further react with $BeCl_2$ to give "beryllium Grignard" compounds:

$$BeMe_2 + BeCl_2 \rightarrow 2MeBeCl$$
(less reactive than the corresponding true Grignard compound)

▮ 7. OXIDES AND PEROXIDES

7.1 Normal Oxides MO

All burn in O_2 to give MO. **BeO** is usually made instead by igniting gelatinous $Be(OH)_2$; the other oxides are usually obtained by **thermal decomposition of the carbonates, sulphates, or nitrates** — all of which are **less thermally stable** than the corresponding Group 1 oxosalts, precisely because the Group 2 metals and their hydroxides are less strongly basic than those of Group 1.

BeO is covalent (4:4 zinc-blende/wurtzite structure); MgO, CaO, SrO, BaO are ionic (6:6 rock-salt/NaCl structure). Basic strength rises steadily: **BeO (amphoteric) < MgO (weakly basic) < CaO < SrO < BaO** (increasingly strongly basic).

MOT / BONDING — Why does MO + H_2O → $M(OH)_2$ happen at all? (mechanistic picture)

The O^{2-} ion in the ionic lattice is an extremely strong **Brønsted base / nucleophile** — far too reactive to survive in water. When water contacts the oxide surface, O^{2-} simply **rips a proton off H_2O**: $O^{2-} + H_2O \rightarrow 2OH^-$. This single proton-transfer step is the real mechanism behind every "basic oxide + water → hydroxide" reaction in this chapter, and it is also exactly why free O^{2-} cannot exist in aqueous solution at all.

$$BeO + 2H^+ \rightarrow Be^{2+} + H_2O$$
$$BeO + 2OH^- + H_2O \rightarrow [Be(OH)_4]^{2-}$$

BeO and MgO are used as furnace-lining **refractories**: very high melting point (BeO ≈2500 °C, MgO ≈2800 °C), negligible vapour pressure, good thermal conductivity, chemical inertness, and electrical insulation.

JEE TRAP

- Simple radius-ratio rules **predict** 8-coordination for SrO and BaO (given the M^{2+}:O^{2-} size ratio), but both are experimentally found to be **6-coordinate** (rock-salt structure) — a known limitation of the radius-ratio approach that advanced questions may test.

7.2 Peroxides

Peroxide stability **increases with cation size** (a larger, less polarizing cation stabilizes the large O_2^{2-} ion better).

- **BaO_2:** $2BaO + O_2 \rightleftharpoons 2BaO_2$ (air passed over BaO at 500 °C) — the classical, most stable Group-2 peroxide.
- **SrO_2:** forms similarly but needs higher pressure and temperature.
- **CaO_2:** not formed by the direct route — instead made as a hydrate by treating $Ca(OH)_2$ with H_2O_2, then dehydrating.
- **MgO_2:** only a crude peroxide obtainable, via H_2O_2 (made by passing H_2O_2 into a suspension of $Mg(OH)_2$, the same method used for CaO_2 from $Ca(OH)_2$). MgO_2 is used as a mild **antiseptic in toothpaste** and as a **bleaching agent**.
- **BeO_2:** no peroxide of beryllium is known.

$$BaO_2 + 2HCl \rightarrow BaCl_2 + H_2O_2$$
$$BaO_2 + H_2SO_4(dil.) \rightarrow BaSO_4\downarrow + H_2O_2$$

MOT / BONDING — Peroxide vs superoxide — bond order from MOT

Peroxide ion $[O–O]^{2-}$ (18 electrons): filling gives $\sigma1s^2\sigma^*1s^2\sigma2s^2\sigma^*2s^2\sigma2p_x^2\pi2p\_y^2\pi2p\_z^2\pi^*2p\_y^2\pi^*2p\_$ → all π^* orbitals fully paired → **bond order = 1**, **diamagnetic** — matches the O–O single bond drawn structurally.

Superoxide ion $[O_2]^-$ (17 electrons): identical filling except the last π^* level holds only **one** electron ($\pi^*2p\_y^2\pi^*2p\_z^1$) → **bond order = 1.5**, **paramagnetic** — this is the standard explanation for why KO_2 is paramagnetic.

All known Group 2 peroxides (BaO_2, SrO_2, etc.) contain the *same* peroxide ion — bond order 1, diamagnetic. No Group 2 superoxide is known to exist.

▮ 8. HYDROXIDES

Basicity and solubility **both increase** down the group. $Be(OH)_2$ is amphoteric; $Mg(OH)_2$ is weakly basic and only sparingly soluble ("milk of magnesia", an antacid); $Ca(OH)_2$, $Sr(OH)_2$, $Ba(OH)_2$ are strong bases.

Hydroxide	Solubility (g/L, ~20 °C)
$Mg(OH)_2$	≈1×10^{-4} (essentially insoluble)
$Ca(OH)_2$	≈2
$Sr(OH)_2$	≈8
$Ba(OH)_2$	≈39

$$Be(OH)_2 + 2OH^- \rightarrow [Be(OH)_4]^{2-}$$
$$Be(OH)_2 + 2HCl + 2H_2O \rightarrow [Be(H_2O)_4]Cl_2$$

NCERT FOCUS — Why does hydroxide solubility increase down the group?

The anion (OH^-) is common across the series, so only the cation's size varies. As M^{2+} grows, **lattice enthalpy falls faster than hydration enthalpy** — so the net dissolution process becomes more favourable and solubility **rises** down the group. This is the exact reverse of what happens for carbonates/sulphates (§8), where hydration enthalpy falls faster than lattice enthalpy, so solubility **falls** down the group.

Master key (memorize this one sentence): *"Which energy — lattice or hydration — falls faster as the cation grows?"* governs every Group 2 solubility trend. For OH^- and F^-, lattice energy falls faster → solubility increases down the group. For CO_3^{2-} and SO_4^{2-}, hydration energy falls faster → solubility decreases down the group.

$Ca(OH)_2$ solution = **lime water**; $Ba(OH)_2$ solution = **baryta water** — both turn milky with CO_2 (precipitate $CaCO_3$/$BaCO_3$), clearing again with excess CO_2 (soluble bicarbonate forms). Baryta water is the more sensitive test — even exhaled breath gives a positive result, whereas lime water needs breath actively bubbled through it.

JEE TRAP

- "Lime water" is a clear **solution**, not a suspension; "milk of lime" is the suspension of excess undissolved $Ca(OH)_2$. Four genuinely distinct terms exist: quicklime (CaO) ≠ slaked lime ($Ca(OH)_2$) ≠ lime water (solution) ≠ milk of lime (suspension).

9. CARBONATES, BICARBONATES & HARDNESS OF WATER

9.1 Carbonates

Solubility **decreases** down the group (hydration energy falls faster than lattice energy for the large CO_3^{2-} ion). All carbonates decompose on heating: $MCO_3 \rightarrow MO + CO_2$. Thermal stability **increases** down the group — smaller, more polarizing cations distort CO_3^{2-} more and decompose more readily.

Carbonate	$BeCO_3$	$MgCO_3$	$CaCO_3$	$SrCO_3$	$BaCO_3$
Decomposition temp (°C)	< 100	540	900	1290	1360

$BeCO_3$ is so unstable it can only be kept under a CO_2 atmosphere, and is structurally unusual — it retains the hydrated $[Be(H_2O)_4]^{2+}$ unit rather than a simple Be^{2+}.

$CaCO_3$ crystal chemistry: calcite (stable form, Ca^{2+} 6-coordinate, matching the CaO-type prediction) vs aragonite (metastable, only ~5 kJ/mol higher in energy but kinetically trapped, Ca^{2+} unusually 9-coordinate).

9.2 Bicarbonates & Hardness of Water

Bicarbonates exist only in solution, never as isolable solids for Group 2:

$$CaCO_3 + CO_2 + H_2O \rightleftharpoons Ca(HCO_3)_2(aq)$$
$$\text{(reverse on heating: } \rightarrow CaCO_3\downarrow + CO_2 + H_2O)$$

This equilibrium, run in reverse by slow evaporation/decomposition, deposits limestone caves' **stalactites** (from the roof) and **stalagmites** (from the floor).

Type	Cause	Removed by
Temporary hardness	$Mg(HCO_3)_2$, $Ca(HCO_3)_2$	Boiling (drives off CO_2); or adding slaked lime, $Ca(HCO_3)_2+Ca(OH)_2 \rightarrow 2CaCO_3+2H_2O$
Permanent hardness	$MgSO_4$, $CaSO_4$ (soluble sulphates)	Not removed by boiling — needs ion exchange, phosphates (calgon), or Na_2CO_3 (lime-soda: $CaSO_4+Na_2CO_3 \rightarrow CaCO_3\downarrow+Na_2SO_4$)

Scum in hard water = insoluble calcium/magnesium stearate, formed when Ca^{2+}/Mg^{2+} react with soap before true lathering can occur.

JEE TRAP

- Excess CO_2 passed into lime water does *not* give more precipitate — it **redissolves** the $CaCO_3$ precipitate as soluble $Ca(HCO_3)_2$, clearing the milkiness. This is a solubility-equilibrium shift, not a contradiction of the earlier precipitation step.

10. SULPHATES AND NITRATES

10.1 Sulphates

Solubility falls sharply: **$BeSO_4$ > $MgSO_4$ ≫ $CaSO_4$ > $SrSO_4$ > $BaSO_4$** (virtually insoluble). Reason: for the small Be^{2+}/Mg^{2+}, hydration enthalpy dominates and overcomes lattice enthalpy, favouring dissolution; for the larger cations, hydration falls off while lattice energy (set mainly by the large SO_4^{2-} ion) stays comparatively high, so solubility collapses.

Salt	Note
Epsom salt, $MgSO_4·7H_2O$	Mild laxative
Alabaster, $CaSO_4·2H_2O$ (fine-grained)	Ornamental; not weatherproof outdoors ($CaSO_4$ is slightly soluble, ~2 g/L)
$BaSO_4$	Insoluble AND opaque to X-rays → "barium meal" for GI-tract imaging

Thermal decomposition $MSO_4 \rightarrow MO + SO_3$ becomes harder as basicity rises: $BeSO_4$ decomposes at 500 °C, $MgSO_4$ at 895 °C, $CaSO_4$ at 1149 °C, $SrSO_4$ at 1374 °C. Heating the sulphates with carbon reduces them to sulphides — $BaSO_4 + 4C \rightarrow BaS + 4CO$ — and most industrial barium compounds are in fact made starting from BaS.

10.2 Perchlorates

Group 2 elements also form perchlorates $M(ClO_4)_2$, which have structures very similar to the sulphates — the ClO_4^- ion is tetrahedral and close in size to SO_4^{2-}. Chemically, however, they differ sharply: perchlorates are strong oxidizing agents, and anhydrous **$Mg(ClO_4)_2$ is used as a drying agent** ("Anhydrone"). It must **never** be used with organic materials, since contact between an organic compound and a strong oxidizer of this kind could cause an explosion.

10.3 Nitrates

Made by dissolving the carbonate/oxide/hydroxide in dilute HNO_3. Hydration of the crystalline nitrate **decreases** down the group: $Mg(NO_3)_2$ crystallizes with $6H_2O$, while $Ba(NO_3)_2$ is anhydrous.

$$2M(NO_3)_2 \rightarrow 2MO + 4NO_2 + O_2$$
(all decompose to the oxide on heating)

Be is unusual in also forming a **basic nitrate [$Be_4O(NO_3)_6$]** in addition to the normal salt. Anhydrous $Be(NO_3)_2$ cannot be obtained by heating the hydrate (decomposition intervenes); instead it is made via liquid N_2O_4:

$$BeCl_2 \xrightarrow{N_2O_4} Be(NO_3)_2·2N_2O_4 \xrightarrow{\text{warm to 50 °C, vacuum}} Be(NO_3)_2 \xrightarrow{125\ °C} [Be_4O(NO_3)_6]$$

JEE TRAP

- Group 2 nitrates decompose straight to the **oxide** (like $LiNO_3$), *not* to a nitrite the way $NaNO_3$/KNO_3 do — do not import the Group-1 heavier-metal nitrite pattern here.

10.4 Insoluble Salts — Qualitative Analysis Summary

A single compact fact worth memorizing as a block: the **sulphates** of calcium, strontium and barium are insoluble, and the **carbonates, oxalates, chromates and fluorides** of the *whole* group are insoluble — this pattern is exactly what is exploited in qualitative inorganic analysis to detect and separate Group 2 cations.

11. ANOMALOUS BEHAVIOUR OF BERYLLIUM

J.D. Lee gives exactly **three** root causes (Group 2, §11.7):

- **1. Extremely small size:** Fajans' rules — small, highly-charged ions tend to form covalent compounds.
- **2. Comparatively high electronegativity** (1.5, vs Mg 1.2, Ca 1.0, Sr 1.0, Ba 0.9) — when Be reacts with another element the electronegativity difference is seldom very large, again favouring covalency. (Even where the difference *is* large, as with F, BeF_2 still behaves as covalent when fused.)
- **3. Only 2s/2p valence orbitals available:** the outer shell can hold a maximum of **8 electrons** (one 2s + three 2p orbitals). Thus Be can form a maximum of **four** conventional 2-electron bonds/coordinate bonds — a maximum coordination number of 4. The later members have larger outer shells and can accommodate more than 8 outer electrons, using *d* orbitals in addition to *s* and *p*, so higher coordination numbers occur (e.g. in the basic acetate).

Consequence: anhydrous Be compounds are predominantly **covalent**, and monomeric BeX_2 molecules *should* be linear — the excited-state Be atom ($1s^2 2s^1 2p^1$, two unpaired electrons) uses sp hybridisation to form two collinear bonds. In fact, such linear monomeric molecules exist only in the **gas phase**: in the solid state, coordination number 4 is always achieved by one of four routes.

MOT / BONDING — Four ways beryllium reaches 4-coordination in the solid/liquid state

1. Coordinate bonds from extra ligands: two ligands with a lone pair use Be's two unfilled orbitals — e.g. two F^- ions coordinate to BeF_2 giving $[BeF_4]^{2-}$; diethyl ether coordinates to $BeCl_2$ giving $BeCl_2(OEt_2)_2$.

2. Polymerisation via bridging halogens: BeX_2 molecules polymerise into chains, e.g. $(BeF_2)_n$, $(BeCl_2)_n$ — each bridging halogen forms one normal covalent bond and donates a lone pair as a coordinate bond.

3. Three-centre two-electron bonding: $(BeMe_2)_n$ has the same chain structure as $(BeCl_2)_n$, but the bonding is instead a 3-centre-2-electron bond covering one Me group and two Be atoms — the same electron-deficient clustering seen in BeH_2.

4. A covalent lattice: BeO and BeS adopt a zinc-blende or wurtzite structure (coordination number 4) rather than the 6:6 rock-salt structure of the heavier oxides.

Hydrolysis of Beryllium Salts

In water, beryllium salts are extensively hydrolysed to give polymeric hydroxo complexes of unknown exact structure — J.D. Lee shows two representative dimeric/trimeric fragments, both built from Be centres bridged by OH groups, each Be remaining 4-coordinate:

$[(HO)_2Be(\mu\text{-}OH)_2Be(OH)_2]^{2-}$-type dimer, and a linear trimer $[(HO)_2Be(\mu\text{-}OH)Be(\mu\text{-}OH)Be(OH)_2]^{2-}$ — both retain tetrahedral 4-coordination at every Be centre via hydroxide bridges.

Adding alkali to these hydrolysed solutions breaks the polymers down to the simple mononuclear **beryllate ion $[Be(OH)_4]^{2-}$** (tetrahedral). Many Be salts instead contain the discrete hydrated ion **$[Be(H_2O)_4]^{2+}$** rather than bare Be^{2+} — stable ionic salts such as $[Be(H_2O)_4]SO_4$, $[Be(H_2O)_4](NO_3)_2$ and $[Be(H_2O)_4]Cl_2$ are known. Forming this hydrated complex effectively spreads the charge over a larger ion, which is why it forms so readily.

Consolidated Reaction Table (J.D. Lee Table 11.9, complete)

Reaction	Comment
$M + 2H_2O \rightarrow M(OH)_2 + H_2$	Be probably reacts with steam (doubtful); Mg reacts with hot water; Ca, Sr, Ba react rapidly with cold water
$M + 2HCl \rightarrow MCl_2 + H_2$	All the metals react with acids liberating hydrogen
$Be + 2NaOH + 2H_2O \rightarrow Na_2[Be(OH)_4] + H_2$	Be is amphoteric
$2M + O_2 \rightarrow 2MO$ (excess dioxygen)	Normal oxide formed by all group members

$Ba + O_2 \rightarrow BaO_2$	Ba also forms the peroxide
$M + H_2 \rightarrow MH_2$	Ionic salt-like hydrides formed at high T by Ca, Sr, Ba
$3M + N_2 \rightarrow M_3N_2$	All form nitrides at high temperatures
$3M + 2P \rightarrow M_3P_2$	All the metals form phosphides at high temperatures
$M + S \rightarrow MS$; $M + Se \rightarrow MSe$; $M + Te \rightarrow MTe$	All the metals form sulphides/selenides/tellurides
$M + F_2 \rightarrow MF_2$; $M + Cl_2 \rightarrow MCl_2$; $M + Br_2 \rightarrow MBr_2$; $M + I_2 \rightarrow MI_2$	All the metals form halides
$M + 2NH_3 \rightarrow M(NH_2)_2 + H_2$ (high temperature)	All the metals form amides at high temperatures

JEE TRAP

- Consequences of Be's anomaly to keep straight: never exceeds CN = 4; oxide/hydroxide uniquely amphoteric; extensive hydrolysis gives acidic solutions (no other Group-2 salt hydrolyses appreciably); rarely more than 4 waters of crystallisation; halides covalent/polymeric/soluble-in-organic-solvents/fuming/subliming/non-conducting when fused; BeH_2 covalent (not saline); Be_2C a methanide ($\rightarrow CH_4$, unlike acetylide $MC_2 \rightarrow C_2H_2$); $BeCO_3$ stable only under CO_2; no flame colour; unusually strong complex-former; amphoteric as the *metal* too (dissolves in NaOH); passivated by concentrated HNO_3 like Al.

The Bigger Picture: Be's Anomaly Is Not a One-Off

Be is just one instance of a completely general pattern called the **"first-element" (or second-period) anomaly**: **every** element that heads a group in the second period — Li, Be, B, C, N, O, F — behaves noticeably differently from the rest of its own group, and for exactly the *same* three root causes seen above for Be: (1) unusually small size, (2) unusually high electronegativity/charge density for the group, and (3) only 2s/2p valence orbitals available, capping the outer shell at 8 electrons and the coordination number at 4 (no accessible *d* orbitals). Li's anomalous behaviour vs. Na/K (its own hydration-driven E° quirk, its nitride formation, its covalent character) is the exact Group-1 mirror of everything you've just learned about Be — recognizing this pattern turns twelve separate "anomaly facts" into one reusable rule.

12. DIAGONAL RELATIONSHIP: BE AND AL

Be^{2+} (ionic radius ≈31 pm) and Al^{3+} (≈50 pm) are not close in raw size, but their **charge densities** (charge per unit surface area) are almost identical (Be^{2+} ≈2.36, Al^{3+} ≈2.50 in the same units) — this is the quantitative root of every diagonal relationship, not just Be–Al.

#	Similarity
1	Neither readily attacked by acids — both passivated by concentrated HNO_3 via a protective oxide film.
2	Both oxides (BeO, Al_2O_3) and hydroxides ($Be(OH)_2$, $Al(OH)_3$) are amphoteric; both dissolve in excess alkali to give beryllate $[Be(OH)_4]^{2-}$ / aluminate $[Al(OH)_4]^-$.
3	Both chlorides are Cl-bridged in the vapour phase ($BeCl_2$ dimer/polymer; Al_2Cl_6 dimer); both soluble in organic solvents, both strong Lewis acids, both used as Friedel–Crafts catalysts.
4	Both form complex fluorides readily: $[BeF_4]^{2-}$, $[AlF_6]^{3-}$ — but note Be maxes out at CN = 4 while Al reaches CN = 6, a genuine point of difference within the similarity.
5	Both carbides hydrolyse to methane: $Be_2C \rightarrow CH_4$; $Al_4C_3 \rightarrow CH_4$ (contrast $CaC_2 \rightarrow C_2H_2$).
6	Both nitrides hydrolyse to NH_3: $Be_3N_2 \rightarrow NH_3$; $AlN \rightarrow NH_3$.
7	Both hydrides are electron-deficient and polymeric with multicentre (3c–2e) bonding: BeH_2 and AlH_3.
8	Standard electrode potentials are close: Be −1.85 V, Al −1.66 V — much closer to each other than Be is to Ca/Sr/Ba (−2.87 to −2.90 V).
9	Be salts are extensively hydrolysed (acidic in solution), as are Al salts.
10	Be salts are among the most soluble known — Al salts show a similar high-solubility tendency.

JEE TRAP

- A diagonal relationship is a set of **similarities, not identity**. Be still differs from Al in maximum coordination number (4 vs 6) and, of course, in oxidation state (+2 vs +3) — never treat Be and Al as interchangeable.

13. DIFFERENCES BETWEEN BERYLLIUM AND THE OTHER GROUP 2 ELEMENTS

A distinct list from the Be–Al diagonal relationship above — this is J.D. Lee's direct within-group comparison (Be vs Mg/Ca/Sr/Ba), given as eight numbered points.

#	Difference
1	Be is very small and has a high charge density, so by Fajans' rules it has a strong tendency to covalency. Thus the melting points of its compounds are lower — BeF_2 melts at 800 °C, whilst the fluorides of the rest of the group melt about 1300 °C. The Be halides are all soluble in organic solvents and hydrolyse in water, rather like the aluminium halides. The other Group 2 halides are ionic.
2	Beryllium hydride is electron-deficient and polymeric, with multicentre bonding, like aluminium hydride.
3	The halides of beryllium are electron-deficient and polymeric, with halogen bridges. $BeCl_2$ usually forms chains but also exists as the dimer; $AlCl_3$ is dimeric.
4	Be forms many complexes — not typical of Groups 1 and 2.
5	Be is amphoteric, liberating H_2 with NaOH and forming beryllates; Al forms aluminates similarly.
6	$Be(OH)_2$, like $Al(OH)_3$, is amphoteric.
7	Be, like Al, is rendered passive by nitric acid.
8	The standard electrode potentials for Be and Al, −1.85 V and −1.66 V respectively, are much closer to each other than the value for Be is to the values for Ca, Sr and Ba (−2.87, −2.89 and −2.91 V respectively).

Three more general points close the chapter's treatment of Be:

- 9. Be salts are extensively hydrolysed.
- 10. Be salts are among the most soluble known.
- 11. Be forms an unusual carbide Be_2C, unlike CaC_2, which yields methane on hydrolysis.

JEE TRAP

- Notice this list (Be vs. the *rest of its own group*) is distinct from — though it overlaps with — the Be–Al diagonal-relationship list in §12. Examiners sometimes ask "which of these is a difference *within* Group 2 vs. a similarity *across* to Group 13" — both framings describe the same underlying facts about Be.

14. COMPOUNDS OF CALCIUM (NCERT SCOPE)

14.1 Calcium Oxide — Quicklime, CaO

$$CaCO_3 \xrightarrow{(1070\text{–}1270\ K,\ \text{rotary kiln})} CaO + CO_2\uparrow$$

(CO_2 removed continuously to drive equilibrium)

- White amorphous solid, m.p. 2870 K; absorbs atmospheric moisture and CO_2 on standing.
- **Slaking:** $CaO + H_2O \rightarrow Ca(OH)_2$ + heat. Slaking with aqueous NaOH gives **soda lime** (a $NaOH/Ca(OH)_2$ mixture, easier to handle than pure NaOH).
- Basic oxide — combines with acidic oxides at high T: $CaO + SiO_2 \rightarrow CaSiO_3$ (this is also literally the base reaction behind ordinary glass-making); $6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$. The same $CaO + SiO_2 \rightarrow CaSiO_3$ reaction is exactly why lime (via $CaCO_3$) works as a metallurgical **flux**: the acidic silica impurity in an iron ore is converted to molten, low-density calcium silicate ("slag"), which floats on the liquid metal and is skimmed off.

NCERT FOCUS — Uses of CaO (NCERT)

- Important raw material for manufacturing cement, and the cheapest form of alkali
- Manufacture of sodium carbonate from caustic soda
- Purification of sugar and manufacture of dyestuffs

J.D. Lee's fuller industrial-use list for lime (CaO) — worth knowing beyond the NCERT three:

- In steelmaking, to remove phosphates and silicates as slag
- By mixing with SiO_2 and alumina or clay, to make cement
- For making glass
- In the lime-soda process, one of the chlor-alkali industry routes converting Na_2CO_3 to NaOH or vice versa
- For "softening" water
- To make slaked lime $Ca(OH)_2$, and to make calcium carbide CaC_2

14.2 Calcium Hydroxide — Slaked Lime, $Ca(OH)_2$

Prepared as above; white amorphous powder, sparingly soluble. Aqueous **solution** = lime water; **suspension** of excess solid = milk of lime.

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3\downarrow + H_2O$$

(milky)

$$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$$

(excess CO_2 clears it)

JEE TRAP

- Lime water does *not* only test for CO_2 — **SO_2 turns it milky too**, by the exactly parallel reaction $Ca(OH)_2 + SO_2 \rightarrow CaSO_3\downarrow + H_2O$, and excess SO_2 likewise clears the milkiness by forming soluble calcium bisulphite: $SO_2 + H_2O + CaSO_3 \rightarrow Ca(HSO_3)_2$. If a question says "a gas turns lime water milky," don't jump straight to CO_2 — S in the +4 state (SO_2) behaves the same way.

$$2Ca(OH)_2 + 2Cl_2 \rightarrow CaCl_2 + Ca(OCl)_2 + 2H_2O$$

(bleaching powder — a simplified representation)

Bleaching powder is often written as $Ca(OCl)_2$ but is really a **mixed salt** — worth noting for JEE Advanced precision rather than treating it as one pure compound.

NCERT FOCUS — Uses of $Ca(OH)_2$ (NCERT)

- Manufacture of mortar, a building material
- Whitewashing, due to its disinfectant nature (hardens by chemical carbonation to $CaCO_3$, not just drying)
- Glass making, tanning industry
- Preparation of bleaching powder
- Purification of sugar

14.3 Calcium Carbonate, $CaCO_3$

$$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$$
$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaCl$$

White fluffy powder, almost insoluble; decomposes at 1200 K to $CaO + CO_2$. Reacts with dilute acid: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$. With H_2SO_4, an insoluble $CaSO_4$ coating can passivate the surface and slow the reaction.

NCERT FOCUS — Uses of $CaCO_3$ (NCERT)

- Building material, as marble
- Manufacture of quicklime
- As a flux (with $MgCO_3$) in extraction of metals such as iron
- Specially precipitated $CaCO_3$ in high-quality paper manufacture
- Antacid, mild abrasive in toothpaste, constituent of chewing gum, filler in cosmetics

14.4 Calcium Sulphate System — Gypsum, Plaster of Paris, Anhydrite

$$2(CaSO_4 \cdot 2H_2O) \xrightarrow{393\ K} 2(CaSO_4 \cdot \tfrac{1}{2}H_2O) + 3H_2O$$
(gypsum → Plaster of Paris)

Staged dehydration on stronger heating: $CaSO_4 \cdot 2H_2O \xrightarrow{150\ °C} CaSO_4 \cdot \tfrac{1}{2}H_2O$ (Plaster of Paris) $\xrightarrow{200\ °C} CaSO_4$ ("dead-burnt" anhydrite) $\xrightarrow{1100\ °C} CaO + SO_3$.

$$CaSO_4 \cdot \tfrac{1}{2}H_2O + 1\tfrac{1}{2}H_2O \rightarrow CaSO_4 \cdot 2H_2O$$
(sets to a hard mass in 5–15 minutes)

"Dead-burnt" plaster (fully anhydrous $CaSO_4$, from overheating past 393 K) has lost the ability to set normally, because rehydration becomes far too slow.

NCERT FOCUS — Uses of Plaster of Paris (NCERT)

- Largest use in the building industry, as plaster
- Immobilising broken bones/sprains
- Dentistry, ornamental work, casts of statues and busts

JEE TRAP

- Gypsum added to Portland cement (2–3%) plays a *different* chemical role than PoP's own setting reaction — it **slows** the setting of the cement's aluminate phase, preventing "flash setting"; it is not there to "set" the way PoP itself sets.
- PoP does *not* set to anhydrous $CaSO_4$ — it rehydrates **back to gypsum** ($CaSO_4 \cdot 2H_2O$). "Dead-burnt" plaster ≠ Plaster of Paris; dead-burnt is the fully anhydrous, non-setting form.

14.5 Cement (Portland Cement)

First made in England in 1824 by **Joseph Aspdin**; named "Portland cement" for its resemblance to limestone quarried on the Isle of Portland. Made by strongly heating limestone with clay to form "cement clinker," then mixing the clinker with 2–3% gypsum.

Component	% by mass
CaO	50–60
SiO_2	20–25
Al_2O_3	5–10
MgO	2–3
Fe_2O_3	1–2
SO_3	1–2

Quality control ratios (NCERT): silica/alumina ratio between 2.5–4; lime/(silica+alumina+iron oxide) ratio as close to 2 as possible.

Cement ingredient	Formula	% present
Dicalcium silicate	Ca_2SiO_4	26
Tricalcium silicate	Ca_3SiO_5	51
Tricalcium aluminate	$Ca_3Al_2O_6$	11

Setting = hydration of these constituents + their rearrangement into a hard interlocking mass; gypsum's role is purely to slow this down enough to avoid flash-setting.

Sorel cement (a different cement entirely, not part of Portland cement chemistry): a paste-like mixture of MgO and $MgCl_2$ that sets to a hard mass on standing; used in dental filling and flooring.

15. BIOLOGICAL IMPORTANCE OF MAGNESIUM AND CALCIUM

An adult human body contains roughly **25 g of Mg** and **1200 g of Ca** (compare only 5 g Fe, 0.06 g Cu) — daily requirement 200–300 mg.

Magnesium

- Cofactor for **all enzymes that utilise ATP** in phosphate transfer (phosphohydrolases, phosphotransferases).
- The central metal ion of **chlorophyll** — essential for light absorption in photosynthesis.
- Essential for transmission of impulses along nerve fibres; concentrated inside cells, like K^+.

Calcium

- ≈99% of body Ca is in bones and teeth as an apatite-type mineral; tooth enamel specifically is **fluorapatite** [$3Ca_3(PO_4)_2 \cdot CaF_2$]. Bone is metabolically active, not inert — roughly 400 mg/day passes through the plasma pool, being continuously dissolved and redeposited.
- Regulates neuromuscular function, interneuronal transmission, cell-membrane integrity, and **blood coagulation**.
- Plasma calcium is held at ~100 mg/L, regulated by **calcitonin and parathyroid hormone**.
- Concentrated *outside* cells (in body fluids), like Na^+; required to trigger **muscle contraction** and to maintain the regular beating of the heart.

16. MASTER JEE TRAP BANK

Thirty commonly-tested false statements, consolidated across sources. Read each as "a JEE option that looks true but isn't."

1. All Group 2 oxides are basic — FALSE: BeO is amphoteric.
2. All Group 2 hydroxides are strong/highly soluble bases — FALSE: $Be(OH)_2$ is amphoteric, $Mg(OH)_2$ is weakly basic and only sparingly soluble.
3. Hydroxide solubility decreases down the group — FALSE: it increases (lattice energy falls faster than hydration energy for OH^-/F^-).
4. Sulphate solubility increases down the group — FALSE: it decreases sharply; $BaSO_4$ is virtually insoluble.
5. Carbonate solubility increases down the group — FALSE: it decreases (opposite mechanism to hydroxides — here hydration energy falls faster than lattice energy).
6. Mg + steam → $Mg(OH)_2$ — FALSE: the steam product is MgO + H_2; $Mg(OH)_2$ forms only with liquid (especially hot) water.
7. Be reacts vigorously with water because its E° is very negative — FALSE: kinetic passivation by the BeO film dominates over thermodynamics.
8. Burning Mg is extinguished by CO_2 — FALSE: Mg reduces CO_2 ($2Mg+CO_2 \rightarrow 2MgO+C$) and keeps burning; never use CO_2 extinguishers on Mg fires.
9. BeH_2 is ionic/saline — FALSE: it is covalent, polymeric, and electron-deficient (3c–2e bonded).
10. CaH_2 is covalent like BeH_2 — FALSE: $CaH_2/SrH_2/BaH_2$ are ionic, containing H^-.
11. $BeCl_2$ is a simple ionic solid — FALSE: it is strongly covalent — chain-polymeric in the solid, dimeric in vapour at moderate T, monomeric (linear) only above ≈1200 K.
12. All Group 2 carbides give ethyne on hydrolysis — FALSE: $Be_2C \rightarrow CH_4$ (methanide); $Mg_2C_3 \rightarrow$ propyne; only $CaC_2/SrC_2/BaC_2$ (true acetylides) → C_2H_2. Also, $BaC_2 + N_2$ gives cyanide $Ba(CN)_2$, not cyanamide, unlike CaC_2.
13. Group 2 nitrates decompose to nitrites on heating — FALSE: all give MO + NO_2 + O_2 (like $LiNO_3$), never a nitrite.

14. Blue liquid-ammonia solutions are coloured by M^{2+} — FALSE: the colour is due to solvated/ammoniated electrons.
15. Be and Mg show characteristic flame colours — FALSE: their valence electrons are too tightly bound to be excited at flame temperature.
16. Lime water is a suspension — FALSE: it is a clear solution; milk of lime is the suspension.
17. Excess CO_2 increases the $CaCO_3$ precipitate — FALSE: excess CO_2 redissolves it as soluble $Ca(HCO_3)_2$, clearing the milkiness.
18. Whitewash hardens purely by water evaporating — FALSE: chemical carbonation of $Ca(OH)_2$ to $CaCO_3$ is the key hardening step.
19. Gypsum and Plaster of Paris are the same hydration state — FALSE: dihydrate (gypsum) vs hemihydrate (PoP).
20. PoP sets to anhydrous $CaSO_4$ — FALSE: it rehydrates back to gypsum ($CaSO_4 \cdot 2H_2O$).
21. Dead-burnt plaster = $CaSO_4 \cdot ½H_2O$ — FALSE: dead-burnt plaster is fully anhydrous $CaSO_4$, from overheating past the PoP stage.
22. More mixing water always gives a stronger PoP set — FALSE: excess water leaves pores after evaporation, generally weakening the set mass.
23. Be resembles only Mg, with no special relation to Al — FALSE: the Be–Al diagonal relationship is extensive and well-documented.
24. Be^{2+} should show a high coordination number because of its high +2 charge — FALSE: its tiny valence shell (only 2s+2p, no d-orbitals) limits it to CN = 4.
25. $BaSO_4$ dissolves readily because Ba^{2+} is a large, "easy" ion — FALSE: it is precisely the reduced hydration enthalpy of the large Ba^{2+} ion that makes $BaSO_4$ so insoluble.
26. E° order can be predicted from ionization enthalpy alone — FALSE: E° is a thermodynamic sum of atomization enthalpy, $IE_1 + IE_2$, and hydration enthalpy.
27. "Lime" is one single substance — FALSE: quicklime (CaO), slaked lime ($Ca(OH)_2$), lime water (solution), and milk of lime (suspension) are four distinct things.
28. Radius-ratio rules always correctly predict ionic coordination number — FALSE: they predict CN = 8 for SrO/BaO, but both are experimentally 6-coordinate.
29. Fluoride and hydroxide solubility trends follow the same logic as carbonate/sulphate — FALSE: F^-/OH^- show the reversed (increasing) solubility trend because lattice energy — not hydration energy — falls faster down the group for these two anions.
30. Anhydrous $MgCl_2$ can be made by simply heating $MgCl_2 \cdot 6H_2O$ in air — FALSE: this causes hydrolysis (→ Mg(OH)Cl → MgO); anhydrous $MgCl_2$ is made industrially via the double salt $MgCl_2 \cdot NH_4Cl \cdot 6H_2O$ route.

Numerical / Integer-Answer JEE Patterns

A recurring JEE Advanced question style asks for a bare integer rather than an MCQ option. These five are drawn directly from the standard Group 2 practice-question banks and are worth memorizing as verified facts, not just methods:

- **Number of chelate rings in $[Ca(EDTA)]^{2-}$ = 5** — hexadentate EDTA wraps a metal ion with 5 five-membered rings, regardless of which metal is chelated.
- **Total electrons in one molecule of Mg_2C_3 = 42** — a straightforward electron-counting exercise (2×12 + 3×6 = 42) that examiners like precisely because it forces you to first get the correct formula (allylide/sesquicarbide, not MgC_2).
- **Number of planes of symmetry in $[BeH_4]^{2-}$ = 6** — a regular tetrahedron (Td point group) has 6 mirror planes, one through each edge-pair.
- **Ratio of water of crystallization, gypsum : Plaster of Paris = 4 : 1** — $CaSO_4 \cdot 2H_2O$ has 2 mol H_2O, $CaSO_4 \cdot ½H_2O$ has 0.5 mol, so 2/0.5 = 4.
- **Number of Group-2/related elements that liberate H_2 with NaOH, from {Be, Al, B, Mg, Ca, Zn, Sn}** — only Be, Al, Zn, Sn are amphoteric enough to do this (Mg, Ca do not react with NaOH; B does not either) — a reminder that amphoterism is the deciding property, not group membership.

▮ 17. QUICK-REVISION SUMMARY SHEET

Solubility trends (down the group)

Species	Trend	Governing factor
Hydroxides $M(OH)_2$	Increases	Lattice energy falls faster than hydration energy
Fluorides MF_2	Increases (mostly)	Same as hydroxides
Carbonates MCO_3	Decreases	Hydration energy falls faster than lattice energy
Sulphates MSO_4	Decreases sharply	Same as carbonates
Halides other than F (Cl, Br, I)	Broadly high throughout	All fairly soluble, ionic for Mg onward

Thermal stability trends (down the group)

Species	Trend	Reason
Carbonates	Increases	Larger, less polarizing cation distorts CO_3^{2-} less
Nitrates	Increases	Same polarization logic
Sulphates	Increases	Same polarization logic
Hydroxides	Increases	Larger cation polarizes OH^- less, resists dehydration

Nature of key compounds

Compound class	Be	Mg	Ca / Sr / Ba
Hydride MH_2	Covalent, polymeric	Covalent, polymeric	Ionic (H^-)
Chloride MCl_2	Covalent, polymeric/dimeric	Largely ionic	Ionic
Oxide MO	Amphoteric, covalent (4:4)	Weakly basic, ionic (6:6)	Strongly basic, ionic (6:6)
Carbide hydrolysis	$\rightarrow CH_4$ (methanide)	$Mg_2C_3 \rightarrow$ propyne	$\rightarrow C_2H_2$ (acetylide)

Flame colours

Be	Mg	Ca	Sr	Ba
None	None	Brick red	Crimson	Apple green

The one sentence that explains every solubility trend

MOT / BONDING — Unifying principle

"As the cation grows down the group, does lattice energy or hydration energy fall faster?" For OH^-/F^- — lattice energy falls faster ⇒ solubility **increases**. For CO_3^{2-}/SO_4^{2-} — hydration energy falls faster ⇒ solubility **decreases**. Every apparently-contradictory Group 2 solubility trend collapses into this single competing-energies framework.

— End of Notes —